SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	Nexgen
Selected Domain	Domian 2
Selected SDG Track (1–17)	2, 8, 12, 13
Project / Solution Name	Orbyte Smart Yield
Name (Leader)	Yeo Jin Hong
Email & Phone (Leader)	kadenyeojinhong@gmail.com / 0143987675
University Affiliation(s)	Xiamen University Malaysia

Number of Team Members 5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Each day, bakeries and food merchants face severe financial losses, operational friction, and inventory inefficiency due to unsold, near-expiry baked goods. Unpredictable daily demand forces store employees to endure exhausting closing hours manually inspecting, paper-logging, and discarding surplus batches, directly eroding merchants' gross margins. Environmentally, these discarded foods decompose in landfills into methane, driving carbon emissions and squandering agricultural resources.

To bridge this operational and ecological gap, **Orbyte Smart Yield** provides a circular surplus food clearance and eco-loyalty platform. Store staff rapidly record remaining inventory through AI-powered natural language POS input, while the system automatically calculates unit costs, markdown pricing, and gross margins protected by automated loss-floor warnings. Once approved via an owner-side dashboard, surplus batches are dispatched to WhatsApp community channels as customizable, allergen-safe mystery boxes. Crucially, the platform tracks real-time carbon emissions averted (CO_2e) and converts ecological savings into green customer points. By transforming commercial food waste into recovered revenue, labor savings, and measurable environmental impact, our solution champions retail efficiency and community-wide sustainability.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Our proposed platform, **Orbyte Smart Yield**, delivers an intelligent food-waste mitigation ecosystem combining operational NLP automation, algorithmic margin safeguarding, and generative marketing, directly leveraging API and database architecture skills gained from SDC Phase 1 workshops:

1. **AI-Powered POS Ingestion & Margin Floor Protection**: Store staff simply type conversational text (e.g., *"12 croissants, 5 chocolate slices"*); our NLP parsing engine immediately identifies items and reconciles stock. The algorithmic backend executes instant financial appraisals—enforcing staple-food inclusions, optimizing markdown pricing, and flagging below-cost discounts with automated yellow margin warnings to safeguard profitability.
2. **Generative Marketing & Community Dispatch**: Integrating the **DeepSeek GenAI API** paired with recommendation rules (recommendation_rules.json), the system crafts persuasive promotional copy with one click. The owner reviews financial recoveries on a dedicated BI dashboard and broadcasts listings directly to local WhatsApp customer groups.
3. **Allergen-Safe Mystery Boxes & Gamified Eco-Points**: Consumers purchase customized mystery boxes filtered by specific allergen preferences (e.g., wheat, milk), redeeming green loyalty points for tangible discounts. Simultaneously, unsold remnants are categorized on the *Rescue* page into donatable items for local charities and inedible food for bioenergy processing.
4. **Standalone Computer-Vision Extension**: To explore automated shelf auditing, our team trained a dedicated **YOLO object detection model** (train_bread). Due to commercial CCTV access restrictions and privacy constraints in retail stores, this vision module is packaged in our repository as an optional, plug-and-play asset rather than a compulsory dependency.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

- **Backend & API Layer:** Developed with FastAPI (Python), exposing asynchronous RESTful endpoints for natural-language POS inventory parsing, algorithmic cost-plus mystery box bundling, gross margin floor protection with automated yellow warnings, and owner BI authorization workflows (router.py).

- **Database & Persistence:** Powered by SQLite managed through an abstraction repository pattern (repository.py) featuring ACID-compliant transaction management, securely housing product catalogs, customer green loyalty accounts, order queues, charity rescue routing, and carbon emission logs (loyalty_ledger.json, order_queue.json).

- **GenAI & Algorithmic Engine:** Integrates DeepSeek GenAI API to dynamically draft contextual promotional copy from structured inventory prompts. A local rule-based recommendation engine utilizes structured JSON rulebases (recommendation_rules.json, recommendation_examples.json) to enforce staple inclusions, allergen exclusions (e.g., wheat, milk), and real-time CO_2e offset conversions.

- **Computer Vision Asset (Standalone):** A fine-tuned YOLO object detection model (train_bread), independently trained and delivered within the codebase as an optional edge-vision module for merchants with open CCTV/RTSP camera access.

- **Frontend & Community Channel:** Built with responsive HTML5, CSS3 (supporting Dark Mode toggle), and modern Vanilla JavaScript, featuring dedicated views for Staff POS, Owner BI, Customer allergen-safe purchases, and Rescue/charity management, integrated with automated WhatsApp dispatch formats.

- **Testing & Build Environment:** Automated with Python test suites (test_fronted.py) and packaged via Yarn for consistent dependency control.

- ***(SDC Sponsor/GenAI Utilization: The project utilizes the DeepSeek GenAI API for automated, localized promotional copywriting and WhatsApp campaign generation, integrated directly into the FastAPI backend pipeline.)***

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

- **Empathize & Define:** Analyzing bakery closing routines revealed dual frictions: staff face exhausting manual sorting under closing deadlines, while owners fear revenue loss. Recognizing third-party CCTV privacy and access constraints, we kept frontline entry zero-friction while packaging our pre-trained YOLO model as a standalone vision asset.

- **Ideate & Prototype**: We decoupled operations into intuitive staff intake and owner-side BI authorization. Phase 1 fully validates: staple-food hard filtering, automated cost-plus mystery box generation, hidden-reward lotteries, and DeepSeek promotional copy dispatch. Crucially, avoided emissions are converted into customer carbon loyalty credits to unlock profile aesthetics and reward redemptions.

- **Testing & Phase 2 Deployment Readiness**: Automated tests (test_fronted.py) confirm REST API stability across core services. For Phase 2 real-world merchant rollout, the architecture transitions to production database transaction management (repository.py), integrating **QR-code scheduled pickup verification**, **no-show penalty freezing**, and **automated surplus liquidation via orphanage donations** to ensure zero food waste.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

*The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess **Innovation and Creativity** as well as **Feasibility of build**. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	• **SDG Target 13.3 (Climate Action):** "Improve education, awareness-raising and human and institutional capacity on climate change mitigation, adaptation, impact reduction and early warning" — addressed by our real-time carbon calculation (kg CO_2e) and consumer carbon loyalty credit ecosystem. • **SDG Target 2.1 / 2.2 (Zero Hunger):** "End hunger and ensure access by all people... to safe, nutritious and sufficient food all year round" — supported by making surplus bakery staples affordable to local communities and redirecting unsold items to orphanages during Phase 2. • **SDG Target 8.2 (Decent Work and Economic Growth):** "Achieve higher levels of economic productivity through diversification, technological upgrading and innovation" — accomplished by recovering lost gross margins for local bakeries and automating exhausting end-of-day staff

Metric Category	Your Response
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	• In standard commercial bakery operations, unsold daily surplus typically accounts for 8% to 12% of total production, resulting in approximately 10–15 kg of safe, edible baked goods discarded per store per night. Closing staff spend 30 to 45 minutes manually inspecting, logging, and discarding surplus batches. Furthermore, based on established FAO/WRAP life-cycle benchmarks, every 1 kg of bread sent to landfills generates approximately 1.4 kg of CO2e through life-cycle emissions and anaerobic decomposition.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	1. Mass of Surplus Food Rescued (kg): Total weight of edible bakery items diverted from disposal into consumer mystery boxes or local charity donations. 2. Avoided Greenhouse Gas Emissions (kg CO2e): Calculated dynamically at 1.4 kg CO2e per kg of food rescued and converted into consumer green loyalty points (e.g., 100 points = RM1). 3. Closing Operational Time Saved (Minutes/Night): Reduction in frontline staff hours spent on end-of-day counting and paper inventory logging. 4. Merchant Revenue Loss Recovery Rate (%): Margin recovered through discounted mystery box clearance versus total stock loss.

Metric Category	Your Response
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation e.g., "20% improvement in computational resource use."*	By the final Phase 2 prototype presentation across pilot bakeries: 1. Achieve a 60% to 75% reduction in daily surplus bakery disposal per pilot store, rescuing 180–250 kg of food per month. 2. Directly prevent an estimated 250–350 kg of CO2e emissions per store monthly. 3. Slash end-of-day inventory and clearance processing time by over 70% (reducing 30–45 minutes of manual sorting down to under 5 minutes using NLP/vision confirmation). 4. Direct 100% of remaining unsold edible food to local orphanages and charities via the Rescue page liquidation pipeline.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

✔️ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

✔️ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

✔️ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

✓ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ✔ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
FastAPI / Pydantic	Code	MIT
Starlette /Uvicorn	Code	BSD-3-Clause
HTTPX / python-dotenv	Code	BSD-3-Clause
Ultralytics YOLOv8	Code / Pre-trained AI Model	AGPL-3.0
Roboflow Universe — Bread Detector	Dataset	CC BY 4.0
OpenCV / NumPy	Code	Apache-2.0 / BSD-3-Clause
PyTorch	Code	BSD-3-Clause
ONNX Runtime Web	Code	MIT
Tailwind CSS	Code	MIT
pytest	Code(testing)	MIT
DeepSeek LLM (API)	AI Model / API	DeepSeek Terms of Service

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ✔ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature] 

[Date] 24-10-26